



US012000821B2

(12) **United States Patent**
Khonsari et al.

(10) **Patent No.:** **US 12,000,821 B2**
(45) **Date of Patent:** **Jun. 4, 2024**

(54) **FLUID MONITORING**

(56) **References Cited**

(71) Applicants: **Michael M. Khonsari**, Baton Rouge,
LA (US); **Lijesh Koottaparambil**,
Baton Rouge, LA (US)

U.S. PATENT DOCUMENTS

8,531,672 B2 * 9/2013 Takahashi G01N 3/56
356/442

(72) Inventors: **Michael M. Khonsari**, Baton Rouge,
LA (US); **Lijesh Koottaparambil**,
Baton Rouge, LA (US)

2019/0234463 A1 * 8/2019 Lugt F16C 33/6633
2021/0389295 A1 * 12/2021 Khonsari F16N 29/00

FOREIGN PATENT DOCUMENTS

(73) Assignee: **Bd. of Supervisors of La. St. Univ. &
A&M College**, Baton Rouge, LA (US)

CN 108008119 A * 5/2018 G01N 33/2888
JP 2020012690 A * 2/2020 G02B 1/115
JP 2020200892 A * 12/2020
JP 2022052591 A * 4/2022
WO WO-2014167503 A1 * 10/2014 G01N 11/00
WO WO-2017007333 A1 * 1/2017 E21B 19/06

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 283 days.

(21) Appl. No.: **17/344,464**

OTHER PUBLICATIONS

(22) Filed: **Jun. 10, 2021**

Rezasoltani, A. Khonsari, M. M. On the Correlation between
Mechanical Degradation of Lubricating Grease and Entropy, Tribology
Letters 60 (1), (2015): 1-4.

(65) **Prior Publication Data**

US 2021/0389295 A1 Dec. 16, 2021

(Continued)

Related U.S. Application Data

(60) Provisional application No. 63/037,421, filed on Jun.
10, 2020.

Primary Examiner — Francis C Gray

(74) *Attorney, Agent, or Firm* — Edel Patents LLC; John
B. Edel

(51) **Int. Cl.**

G01N 33/28 (2006.01)

F16N 29/00 (2006.01)

G01N 33/30 (2006.01)

(52) **U.S. Cl.**

CPC **G01N 33/2888** (2013.01); **F16N 29/00**
(2013.01); **G01N 33/30** (2013.01); **F16N**
2200/00 (2013.01)

(57)

ABSTRACT

Methods of evaluating mixing are disclosed relating to the
preparation and mixing of materials. Materials prepared may
transition from heterogeneous mixtures to homogenous mix-
tures while the current of a motor is monitored. Quantities
relating to the rate of change of power or the rate of change
of entropy may be calculated in evaluating the sufficiency of
mixing. Methods of evaluating lubricant degradation are
also disclosed.

(58) **Field of Classification Search**

CPC G01N 33/2888; G01N 33/30; F16N 29/00;
F16N 2200/00

See application file for complete search history.

19 Claims, 10 Drawing Sheets

